# HiPPO Functional Memory: Coordinate-Queryable Attention for Irregular Time Series

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## Abstract

Irregular time series require models to answer temporal queries at arbitrary coordinates, yet existing approaches typically retain observation-dependent representations that must be repeatedly resolved for each query. We introduce HiPPO Functional Memory (HiPPO-FM), a functional-memory attention operator that compiles an irregular history into a reusable coordinate-addressable state. Instead of storing observation-indexed tokens or latent trajectories, HiPPO-FM writes observations into a fixed-rank functional space and performs direct coordinate evaluation through matched functional geometry. Kernel-smoothed HiPPO coordinates and Gram-based readout define a shared write/read mechanism, while observed-support self-attention enables content-dependent representations before compression. We characterize the induced finite-dimensional query space and derive an error decomposition separating functional geometry approximation from learned-memory mismatch. Across continuous-coordinate reconstruction, irregular representation learning, real-world extrapolation, and event-sequence prediction, HiPPO-FM consistently improves strong irregular time-series baselines, achieving strong reconstruction accuracy on Spiral, competitive performance across 20 UEA datasets, improved real-world extrapolation, and superior results on several irregular event benchmarks. After encoding, the compact functional state can be reused across different query sets without reprocessing the observation history.

## 1 Introduction

An irregular time series is observed at times determined by data collection and queried at times specified by the task. Clinical measurements and environmental observations arrive at uneven intervals, while downstream analysis may require interpolation at missing times, extrapolation beyond the observed interval, or evaluation at a different temporal resolution (Rubanova et al., 2019; Schirmer et al., 2022). These requests operate on the same history at different query coordinates. The representation retained from that history therefore determines how new temporal coordinates can be evaluated.

Existing approaches handle irregularity through continuous latent dynamics, time-aware attention, or structured representations of the observed samples. Latent ODEs and Neural CDEs model continuous hidden trajectories (Rubanova et al., 2019; Kidger et al., 2020); mTAN and ContiFormer introduce continuous-time structure into attention (Shukla & Marlin, 2021; Chen et al., 2023); and patch- or graph-based methods reorganize irregular observations before prediction (Zhang et al., 2024; Liu et al., 2026a). These approaches substantially improve how irregular histories are processed. In observation-indexed readouts, however, a requested coordinate is still resolved through a representation tied to the observed history. The form of this retained representation determines both how temporal queries access the history and what information must remain available for subsequent prediction.

**How can an irregular history be compressed into a function-valued memory that supports direct evaluation at arbitrary temporal coordinates?** We address this question by giving the retained state an explicit continuous coordinate semantics. Observations are written into a shared functional coordinate system, and requested times are evaluated in the same system. Content interaction occurs